

E-COMMERCE STORE DATASET ANALYSIS – SUMMARY

1.OVERVIEW

This analysis is based on an E-commerce retail dataset containing customer, product, store, and sales transaction data across multiple U.S. regions. The dataset includes key business dimensions such as customer demographics, product categories, store formats, and payment methods. The objective of the analysis is to understand sales performance, customer behavior, product demand, and store efficiency using Excel-based data processing and pivot table insights.

The data reveals a diversified retail structure with customers segmented by loyalty levels, products grouped into categories, and stores distributed across North, South, East, and West regions. Sales transactions capture essential metrics such as quantity, unit price, discounts, and payment methods.

2. OBJECTIVE

The primary objectives of this analysis are:

- To evaluate overall business performance through revenue and sales trends
- To identify top-performing customers, products, and stores
- To analyze customer segmentation based on loyalty, gender, and geography
- To understand product demand and category-wise performance
- To assess store and regional efficiency in generating revenue
- To study payment method preferences and discount effectiveness

3. PROBLEM STATEMENT

The business aims to optimize its sales performance and operational efficiency but faces several challenges:

- Uneven revenue distribution across stores and regions (e.g., Store 8 performing best while Store 5 underperforms)

- Inconsistent product data, including missing cost and stock values requiring imputation
- Weak performance in certain regions (notably the West region)
- Over-reliance on specific payment methods (PayPal dominates, while Credit Card and COD are underutilized)
- Need for better targeting of high-performing products for inventory and promotional strategies

The challenge is to convert raw transactional data into actionable insights that support better decision-making in marketing, inventory planning, pricing strategies, and store expansion.

4. DATA DESCRIPTION

- **Customer Table:** Contains demographic details, loyalty levels, and customer segmentation
- **Product Table:** Includes product categories, pricing, brand, and stock information
- **Store Table:** Represents store locations, regions, and store types (Flagship, Outlet, Online)
- **Sales Table:** Contains transaction-level data such as quantity, discount, payment method, and revenue

5. DATA CLEANING AND PREPARATION IN POWER QUERY

- Handled missing values (AVERAGEIF, MODE)
- Removed duplicates
- Corrected data types

6. DATA TRANSFORMATION

- Created new fields (customer Category, Updated stock, Updated cost)
- Used Excel formulas (IF, VLOOKUP SUMIF, AVERAGEIFS, IFS)
- Discount amount and Total amount are calculated using excel formulas

Discount amount = [@[UnitPrice]] * [@[Discount]]

Total amount = [@[UnitPrice]]-[@[Discount Amount]]

6. Exploratory Data Analysis (EDA)

- Basic analysis using filters and sorting
- Understanding trends and patterns
- Identifying outliers or anomalies

7.PIVOT TABLE ANALYSIS

- Customer count by Loyalty level, State, Gender
- Revenue of quantity by products, Category
- Store count by Region and Storetype
- Revenue by region, storetype, Store Id, Payment Type

8. DATA VISUALIZATION (CHARTS & DASHBOARDS)

- Bar charts (Revenue by store id)
- Pie charts (payment methods, categories)
- Line charts (revenue by region)
- Column charts (Customer count by state and loyalty level)
- Pie chart – Customer count by gender
- Dashboard creation with slicers for Category and Loyalty levels, region and store id.

9. KEY INSIGHTS

- Platinum customers represent the most valuable segment
- Sports category dominates in product sales and revenue
- Prod37, Prod39, and Prod34 are top-performing products
- Online stores generate higher revenue compared to physical formats
- Store 8 is the highest revenue-generating store
- West region shows comparatively lower performance and requires improvement
- PayPal is the most frequently used payment method
- Lower discount rates (around 1%) contribute to higher revenue generation

10.BUSINESS RECOMMENDATIONS

- Improve low-performing regions/stores
- Focus on top-selling products
- Optimize discount strategy
- Improve payment method adoption
- Inventory planning suggestions

11. CONCLUSION

This analysis highlights critical business patterns that can help improve decision-making across marketing, sales, and operations. By focusing on high-performing products, optimizing underperforming regions, and diversifying payment and promotional strategies, the company can enhance overall profitability and customer satisfaction.

